

EXPERIMENT : 13Macros in C

- 1). Write a program to define multiplication macros to perform arithmetic functions.

```
#include <stdio.h>
```

```
#define ADD(a, b) ((a) + (b))  
#define SUBTRACT(a, b) ((a) - (b))  
#define MULTIPLY(a, b) ((a) * (b))  
#define DIVIDE(a, b) ((b) != 0 ? ((a) / (b)) : 0)  
#define MOD(a, b) ((a) % (b))
```

```
int main () {
```

```
    int x = 20, y = 3;
```

```
    printf ("Addition (%d + %d) = %d\n", x, y,  
           ADD(x, y));
```

```
    printf ("Subtraction (%d - %d) = %d\n", x, y,  
           SUBTRACT(x, y));
```

```
    printf ("Multiplication (%d * %d) = %d\n", x, y,  
           MULTIPLY(x, y));
```

```
    printf ("Division (%d / %d) = %d\n", x, y,  
           DIVIDE(x, y));
```

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```
printf ("Modulus (%d %% %d) = %d\n",  
        x, y, MOD (x, y));
```

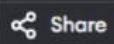
```
return 0;
```

```
}
```

Output :-

Addition ($20 + 3$)	=	23
Subtraction ($20 - 3$)	=	17
Multiplication ($20 * 3$)	=	60
Division ($20 / 3$)	=	6
Modulus ($20 \% 3$)	=	2

Teacher's Signature: _____



```
1 #include <stdio.h>
2
3 #define ADD(a, b)      ((a) + (b))
4 #define SUBTRACT(a, b) ((a) - (b))
5 #define MULTIPLY(a, b) ((a) * (b))
6 #define DIVIDE(a, b)   ((b) != 0 ? ((a) / (b)) : 0)
7 #define MOD(a, b)      ((a) % (b))
8
9 int main() {
10
11     int x = 20, y = 3;
12
13     printf("Addition (%d + %d)      = %d\n", x, y, ADD(x, y));
14     printf("Subtraction (%d - %d)   = %d\n", x, y, SUBTRACT(x, y));
15     printf("Multiplication (%d * %d) = %d\n", x, y, MULTIPLY(x, y));
16     printf("Division (%d / %d)      = %d\n", x, y, DIVIDE(x, y));
17     printf("Modulus (%d %% %d)      = %d\n", x, y, MOD(x, y));
18
19     return 0;
20 }
21
```

```
Addition (20 + 3)      = 23
Subtraction (20 - 3)   = 17
Multiplication (20 * 3) = 60
Division (20 / 3)      = 6
Modulus (20 % 3)       = 2
```

```
=== Code Execution Successful ===
```